2

RS-232 fundamentals

Objectives

When you have completed study of this chapter, you will be able to:

- List the main features of the RS-232 standard
- Fix the following problems:
  - Incorrect RS-232 cabling
  - Male/female D-type connector confusion
  - Wrong DTE/DCE configuration
  - Handshaking
  - Incorrect signaling voltages
  - Excessive electrical noise
  - Isolation

2.1 RS-232 Interface standard (CCITT V.24 Interface standard)

The RS-232 interface standard was developed for the single purpose of interfacing *data terminal equipment* (DTE) and data *circuit terminating equipment* (DCE) employing serial binary data interchange. In particular, RS-232 was developed for interfacing data terminals to modems.

The RS-232 interface standard was issued in the USA in 1969 by the engineering department of the RS. Almost immediately, minor revisions were made and RS-232C was issued. RS-232 was originally named RS-232, (Recommended Standard), which is still in popular usage. The prefix 'RS' was superseded by 'EIA/TIA' in 1988. The current revision is EIA/TIA-232E (1991), which brings it into line with the international standards ITU V.24, ITU V.28 and ISO-2110.

Poor interpretation of RS-232 has been responsible for many problems in interfacing equipment from different manufacturers. This had led some users to dispute as to whether it is a 'standard.' It should be emphasized that RS-232 and other related RS standards

define the electrical and mechanical details of the interface (layer 1 of the OSI model) and do not define a protocol.

The RS-232 interface standard specifies the method of connection of two devices – the DTE and DCE. DTE refers to data terminal equipment, for example, a computer or a printer. A DTE device communicates with a DCE device. DCE, on the other hand, refers to data communications equipment such as a modem. DCE equipment is now also called data circuit-terminating equipment in EIA/TIA-232E. A DCE device receives data from the DTE and retransmits to another DCE device via a data communications link such as a telephone link.

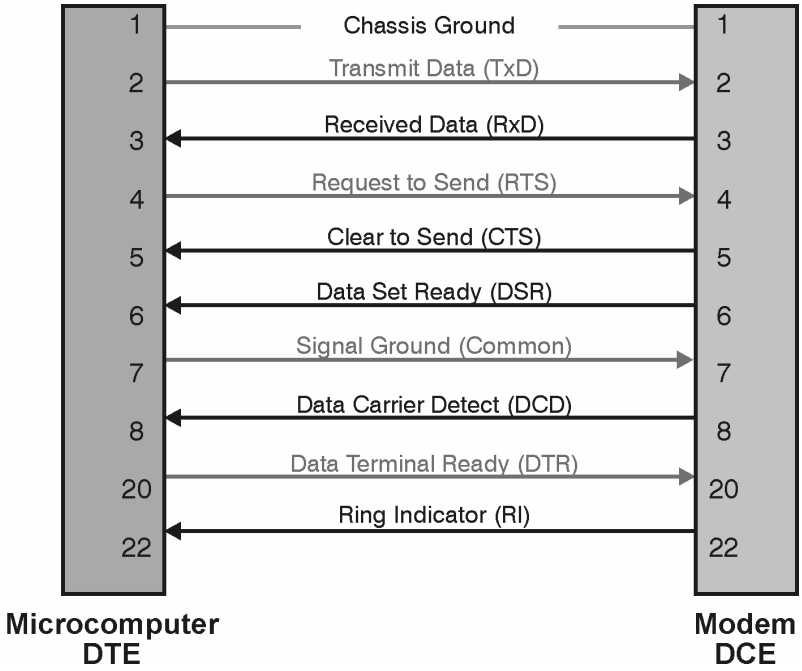


Figure 2.1
Connections between the DTE and the DCE using DB-25 connectors

2.1.1 The major elements of RS-232

The RS-232 standard consists of three major parts, which define:

- Electrical signal characteristics
- Mechanical characteristics of the interface
- Functional description of the interchange circuits

Electrical signal characteristics

RS-232 defines electrical signal characteristics such as the voltage levels and grounding characteristics of the interchange signals and associated circuitry for an unbalanced system.

The RS-232 transmitter is required to produce voltages in the range +/1 5 to +/– 25 V as follows:

- Logic 1: –5 V to –25 V
- Logic 0: +5 V to +25 V
- Undefined logic level: +5 V to –5 V

At the RS-232 receiver, the following voltage levels are defined:

- Logic 1: –3 V to –25 V
- Logic 0: +3 V to +25 V
- Undefined logic level: –3 V to +3 V

Note: The RS-232 transmitter requires a slightly higher voltage to overcome voltage drop along the line.

The voltage levels associated with a microprocessor are typically 0 V to +5 V for *Transistor-Transistor Logic* (TTL). A line driver is required at the transmitting end to adjust the voltage to the correct level for the communications link. Similarly, a line receiver is required at the receiving end to translate the voltage on the communications link to the correct TTL voltages for interfacing to a microprocessor. Despite the bipolar input voltage, TTL compatible RS-232 receivers operate on a single +5 V supply.

Modern PC power supplies usually have a standard +12 V output that could be used for the line driver.

The control or 'handshaking' lines have the same range of voltages as the transmission of logic 0 and logic 1, except that they are of opposite polarity. This means that:

- A control line asserted or made active by the transmitting device has a voltage range of +5 V to +25 V. The receiving device connected to this control line allows a voltage range of +3 V to +25 V
- A control line inhibited or made inactive by the transmitting device has a voltage range of –5 V to –25 V. The receiving device of this control line allows a voltage range of –3 V to –25 V

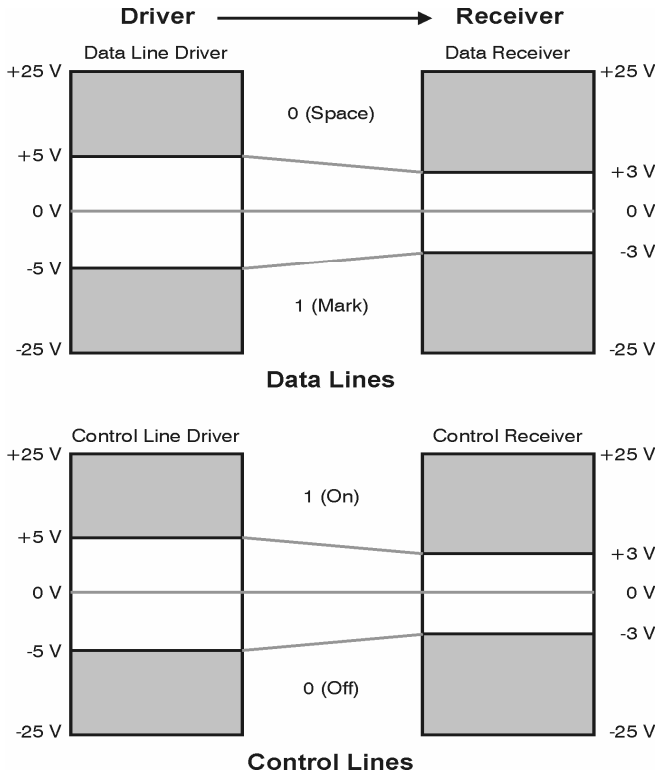


Figure 2.2
Voltage levels for RS-232

At the receiving end, a line receiver is necessary in each data and control line to reduce the voltage level to the 0 V and +5 V logic levels required by the internal electronics.

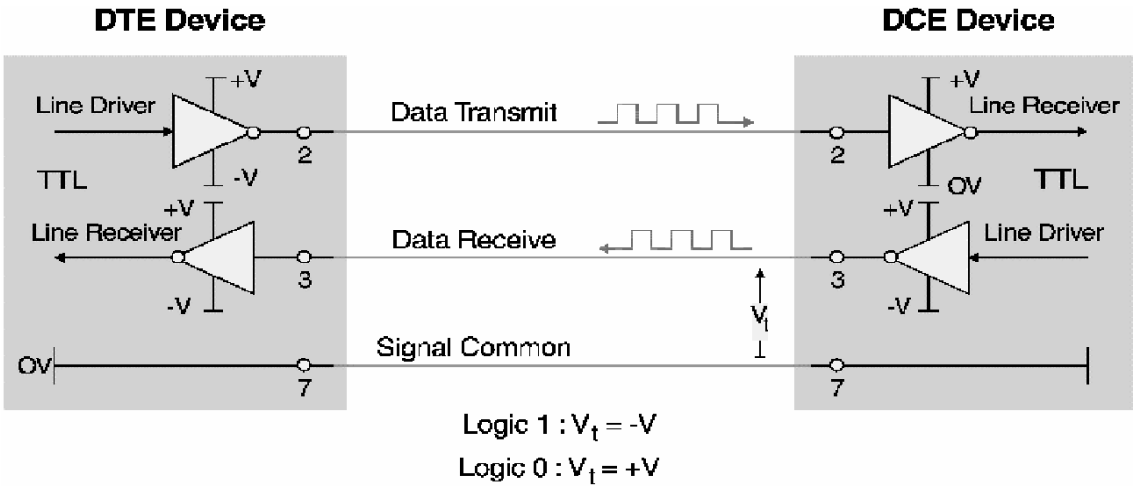


Figure 2.3
RS-232 transmitters and receivers

The RS-232 standard defines 25 electrical connections. The electrical connections are divided into four groups viz:

- Data lines
- Control lines
- Timing lines
- Special secondary functions

Data lines are used for the transfer of data. Data flow is designated from the perspective of the DTE interface. The transmit line, on which the DTE transmits and the DCE receives, is associated with pin 2 at the DTE end and pin 2 at the DCE end for a DB-25 connector. These allocations are reversed for DB-9 connectors. The receive line, on which the DTE receives, and the DCE transmits, is associated with pin 3 at the DTE end and pin 3 at the DCE end. Pin 7 is the common return line for the transmit and receive data lines.

Control lines are used for interactive device control, which is commonly known as hardware handshaking. They regulate the way in which data flows across the interface.

The four most commonly used control lines are:

- RTS: Request to send
- CTS: Clear to send
- DSR: Data set ready (or DCE ready in RS-232D/E)
- DTR: Data terminal ready (or DTE ready in RS-232D/E)

It is important to remember that with the handshaking lines, the enabled state means a positive voltage and the disabled state means a negative voltage.

Hardware handshaking is the cause of most interfacing problems. Manufacturers sometimes omit control lines from their RS-232 equipment or assign unusual applications to them. Consequently, many applications do not use hardware handshaking but, instead, use only the three data lines (transmit, receive and signal common ground) with some form of software handshaking. The control of data flow is then part of the application

program. Most of the systems encountered in data communications for instrumentation and control use some sort of software-based protocol in preference to hardware handshaking.

There is a relationship between the allowable speed of data transmission and the length of the cable connecting the two devices on the RS-232 interface. As the speed of data transmission increases, the quality of the signal transition from one voltage level to another, for example, from –25V to +25 V, becomes increasingly dependent on the capacitance and inductance of the cable.

The rate at which voltage can 'slew' from one logic level to another depends mainly on the cable capacitance and the capacitance increases with cable length. The length of the cable is limited by the number of data errors acceptable during transmission. The RS-232 D&E standard specifies the limit of total cable capacitance to be 2500 pF. With typical cable capacitance having improved from around 160 pF/m to only 50 pF/m in recent years, the maximum cable length has extended from around 15 meters (50 feet) to about 50 meters (166 feet).

The common data transmission rates used with RS-232 are 110, 300, 600, 1200, 2400, 4800, 9600 and 19200 bps. For short distances, however, transmission rates of 38400, 57600 and 115200 can also be used. Based on field tests, table 2.1 shows the practical relationship between selected baud rates and maximum allowable cable length, indicating that much longer cable lengths are possible at lower baud rates. Note that the achievable speed depends on the transmitter voltages, cable capacitance (as discussed above) as well as the noise environment.

Baud Rate	Cable Length (meters)
110	850
300	800
600	700
1200	500
2400	200
4800	100
9600	70
19200	50
115K	20

Table 2.1
Demonstrated maximum cable lengths with RS-232 interface

Mechanical characteristics of the interface

RS-232 defines the mechanical characteristics of the interface between the DTE and the DCE. This dictates that the interface must consist of a plug and socket and that the socket will normally be on the DCE.

Although not specified by RS-232C, the DB-25 connector (25 pin, D-type) is closely associated with RS-232 and is the *de facto* standard with revision D. Revision E formally specifies a new connector in the 26-pin alternative connector (known as the ALT A connector). This connector supports all 25 signals associated with RS-232. ALT A is physically smaller than the DB-25 and satisfies the demand for a smaller connector suitable for modern computers. Pin 26 is not currently used. On some RS-232 compatible equipment, where little or no handshaking is required, the DB-9 connector (9 pin, D-type) is common. This practice originated when IBM decided to make a combined

serial/parallel adapter for the AT&T personal computer. A small connector format was needed to allow both interfaces to fit onto the back of a standard ISA interface card. Subsequently, the DB-9 connector has also became an industry standard to reduce the wastage of pins. The pin allocations commonly used with the DB-9 and DB-25 connectors for the RS-232 interface are shown in table 2.2. The pin allocation for the DB-9 connector is not the same as the DB-25 and often traps the unwary.

The data pins of DB-9 IBM connector are allocated as follows:

- Data transmit pin 3
- Data receive pin 2
- Signal common pin 5

Pin No. DTE	DB-9 Connector IBM Assignment	DB-25 Connector EIA-232 Pin Assignment	DB-25 Connector EIA-530 Pin Assignment
1	Received Line Signal	Shield	Shield
2	Received Data	Transmitted Data	Transmitted Data (A)
3	Transmitted Data	Received Data	Received Data (A)
4	DTE Ready	Request to Send	Request to Send (A)
5	Signal/Common Ground	Clear to Send	Clear to Send (A)
6	DCE Ready	DCE Ready	DCE Ready (A)
7	Request to Send	Signal/Common Ground	Signal/Common Ground
8	Clear to Send	Received Line Signal	Received Line Signal (A)
9	Ring Indicator	+Voltage (testing)	Receiver Signal DCE Element Timing (B)
10		-Voltage (testing)	Received Line (B)
11		Unassigned	Transmitter Signal DTE Element Timing (B)
12		Sec Received Line Signal Detector/Data Signal	Transmitter Signal DCE Element Timing
13		Sec Clear to Send	Clear to Send (B)
14		Sec Transmitted Data	Transmitted Data (B)
15		Transmitter Signal DCE Element Timing	Transmitter Signal DCE Element Timing (A)
16		Sec Received Data	Received Data (B)
17		Receiver Signal DCE Element Timing	Receiver Signal DCE Element Timing (A)
18		Local Loopback	Local Loopback
19		Sec Request to Send	Request to Send (B)
20		DTE Ready	DTE Ready (A)
21		Remote Loopback/Signal Quality Detector	Remote Loopback
22		Ring Indicator	DCE Ready (B)
23		Data Signal Rate	DTE Ready (B)
24		Transmit Signal DTE Element Timing	Transmitter Signal DTE Element Timing (A)
25		Test Mode	Test Mode

Table 2.2
Common DB-9 and DB-25 pin assignments for RS-232 and EIA/TIA-530 (often used for RS-422 and RS-485)

Functional description of the interchange circuits

RS-232 defines the function of the data, timing and control signals used at the interface of the DTE and DCE. However, very few of the definitions are relevant to applications for data communications for instrumentation and control.

The circuit functions are defined with reference to the DTE as follows:

- **Protective ground (shield)**
  The protective ground ensures that the DTE and DCE chassis are at equal potentials (remember that this protective ground could cause problems with circulating earth currents)
- **Transmitted data (TxD)**
  This line carries serial data from the DTE to the corresponding pin on the DCE. The line is held at a negative voltage during periods of line idle
- **Received data (RxD)**
  This line carries serial data from the DCE to the corresponding pin on the DTE
- **Request To Send (RTS)**
  (RTS) is the request to send hardware control line. This line is placed active (+V) when the DTE requests permission to send data. The DCE then activates (+V) the CTS (Clear To Send) for hardware data flow control
- **Clear To Send (CTS)**
  When a half-duplex modem is receiving, the DTE keeps RTS inhibited. When it is the DTE's turn to transmit, it advises the modem by asserting the RTS pin. When the modem asserts the CTS, it informs the DTE that it is now safe to send data
- **DCE ready**
  Formerly called data set ready (DSR). The DTE ready line is an indication from the DCE to the DTE that the modem is ready
- **Signal ground (common)**
  This is the common return line for all the data transmit and receive signals and all other circuits in the interface. The connection between the two ends is always made
- **Data Carrier Detect (DCD)**
  This is also called the received line signal detector. It is asserted by the modem when it receives a remote carrier and remains asserted for the duration of the link
- **DTE ready (data terminal ready)**
  Formerly called data terminal ready (DTR). DTE ready enables but does not cause, the modem to switch onto the line. In originate mode, DTE ready must be asserted in order to auto dial. In answer mode, DTE ready must be asserted to auto answer
- **Ring indicator**
  This pin is asserted during a ring voltage on the line
- **Data Signal Rate Selector (DSRS)**
  When two data rates are possible, the higher is selected by asserting DSRS; however, this line is not used much these days

Pin No.	CCITT No.	Circuit	Description	Circuit Direction
1	-	-	Shield	To DCE
2	103	BA	Transmitted Data	To DCE
3	104	BB	Received Data	From DCE
4	105/133	CA/CJ	Request to Send/Ready for Receiving	To DCE
5	106	CB	Clear to Send	From DCE
6	107	CC	DCE Ready	From DCE
7	102	AB	Signal Common	-
8	109	CF	Received Line Signal Detector	From DCE
9	-	-	Reserved for Testing	-
10	-	-	Reserved for Testing	-
11	126	See Note	Unassigned	-
12	122/112	SCF/CI	Secondary Received Line Signal Detector/ Data Signal Rate Selector	From DCE
13	121	SCB	Secondary Clear to Send	From DCE
14	118	SBA	Secondary Transmitted Data	To DCE
15	114	DB	Transmitter Signal Element Timing (DTE Source)	From DCE
16	119	SBB	Secondary Received Data	From DCE
18	141	LL	Local Loopback	To DCE
19	120	SCA	Secondary Request to Send	To DCE
20	108/112	CD	DTE Ready	To DCE
21	140/110	RL/CG	Remote Loopback/Signal Quality Detector	T/F DCE
22	125	CE	Ring Indicator	From DCE
23	111/112	CH/CI	Data Signal Rate Selector (DTE/DCE Source)	T/F DCE
24	113	DA	Transmit Signal Element Timing (DTE Source)	To DCE
25	142	TM	Test Mode	From DCE
26		None	(Alt A Connector) No Connection at this Time	

Table 2.3
ITU-T V24 pin assignment (ISO 2110)

2.2 Half-duplex operation of the RS-232 interface

The following description of one particular operation of the RS-232 interface is based on half-duplex data interchange. The description encompasses the more generally used full duplex operation.

Figure 2.4 shows the operation with the initiating user terminal, DTE, and its associated modem DCE on the left of the diagram and the remote computer and its modem on the right.

The following sequence of steps occurs when a user sends information over a telephone link to a remote modem and computer:

- The initiating user manually dials the number of the remote computer
- The receiving modem asserts the *Ring Indicator* line (RI) in a pulsed ON/OFF fashion reflecting the ringing tone. The remote computer already has its *Data Terminal Ready* (DTR) line asserted to indicate that it is ready to receive calls. Alternatively, the remote computer may assert the DTR line after a few rings. The remote computer then sets its *Request To Send* (RTS) line to ON

- The receiving modem answers the phone and transmits a carrier signal to the initiating end. It asserts the DCE ready line after a few seconds
- The initiating modem asserts the *data carrier detect* (DCD) line. The initiating
  terminal asserts its DTR, if it is not already high. The modem responds by asserting its DTE ready line
- The receiving modem asserts its *clear to send* (CTS) line, which permits the transfer of data from the remote computer to the initiating side
- Data is transferred from the receiving DTE (transmitted data) to the receiving modem. The receiving remote computer then transmits a short message to indicate to the originating terminal that it can proceed with the data transfer. The originating modem transmits the data to the originating terminal
- The receiving terminal sets its request to send (RTS) line to OFF The receiving modem then sets its *clear to send* (CTS) line to OFF
- The receiving modem switches its carrier signal OFF
- The originating terminal detects that the *data carrier detect* (DCD) signal has been switched OFF on the originating modem and switches its RTS line to the ON state. The originating modem indicates that transmission can proceed by setting its CTS line to ON
- Transmission of data proceeds from the originating terminal to the remote computer
- When the interchange is complete, both carriers are switched OFF and in many cases; the DTR is set to OFF. This means that the CTS, RTS and DCE ready lines are set to OFF

Full duplex operation requires that transmission and reception occur simultaneously. In this case, there is no RTS/CTS interaction at either end. The RTS line and CTS line are left ON with a carrier to the remote computer.

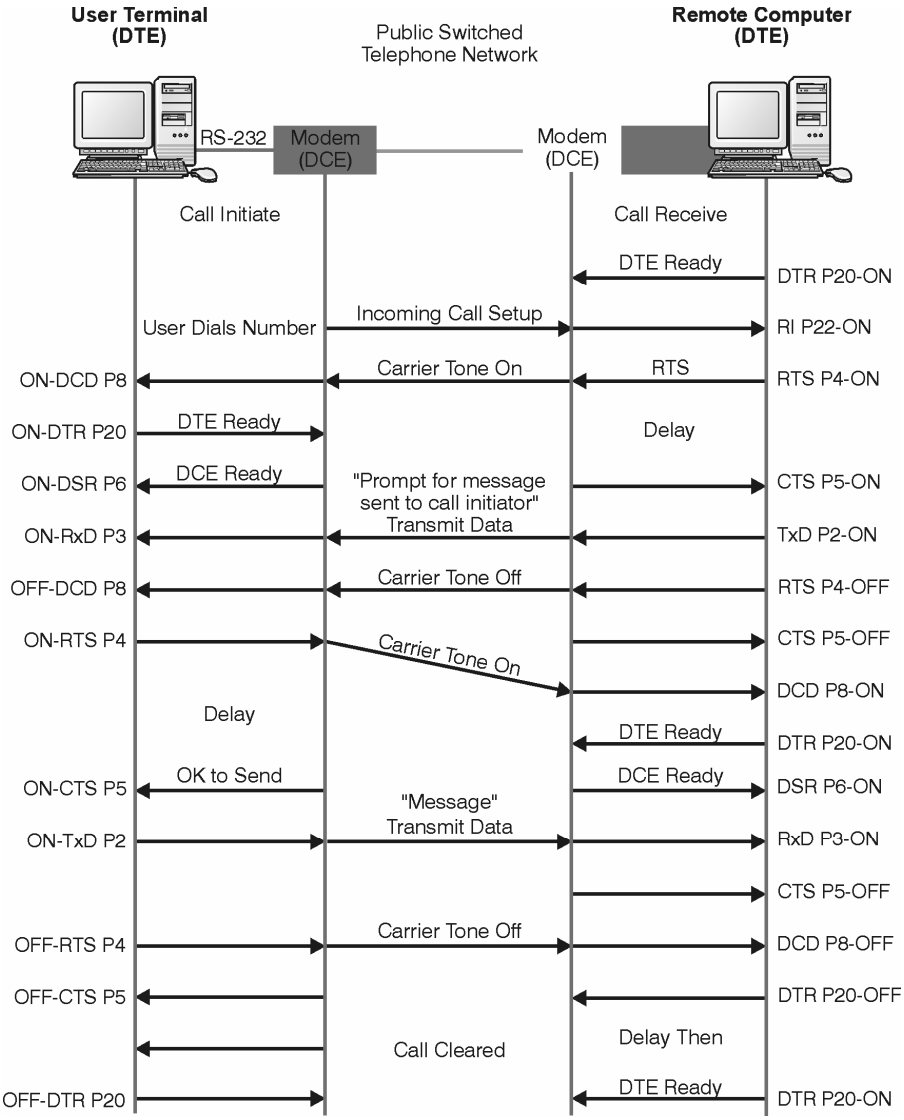


Figure 2.4
Half duplex operational sequence of RS-232

2.3 Summary of EIA/TIA-232 revisions

A summary of the main differences between RS-232 revisions, C, D and E are discussed below.

2.3.1 Revision D – RS-232D

The 25 pin D type connector was formally specified. In revision C, reference was made to the D type connector in the appendices and a disclaimer was included revealing that it was not intended to be part of the standard; however, it was treated as the *de facto* standard.

The voltage ranges for the control and data signals were extended to a maximum limit of 25 V from the previously specified 15 V in revision C.

The 15 meter (50 foot) distance constraint, implicitly imposed to comply with circuit capacitance, was replaced by 'circuit capacitance shall not exceed 2500 pF' (Standard RS-232 cable has a capacitance of 50 pF/ft.).

2.3.2 Revision E – RS-232E

Revision E formally specifies the new 26 pin alternative connector, the ALT A connector. This connector supports all 25 signals associated with RS-232, unlike the 9-pin connector, which has become associated with RS-232 in recent years. Pin 26 is currently not used. The technical changes implemented by RS-232E do not present compatibility problems with equipment confirming to previous versions of RS-232.

This revision brings the RS-232 standard into line with international standards CCITT V.24, V.28 and ISO 2110.

2.4 Limitations

In spite of its popularity and extensive use, it should be remembered that the RS-232 interface standard was originally developed for interfacing data terminals to modems. In the context of modern requirements, RS-232 has several weaknesses. Most have arisen as a result of the increased requirements for interfacing other devices such as PCs, digital instrumentation, digital variable speed drives, power system monitors and other peripheral devices in industrial plants.

The main limitations of RS-232 when used for the communications of instrumentation and control equipment in an industrial environment are:

- The point-to-point restriction, a severe limitation when several 'smart' instruments are used
- The distance limitation of 15 meters (50 feet) end-to-end, too short for most control systems
- The 20 Kbps rate, too slow for many applications
- The –3 to –25 V and +3 to +25 V signal levels, not directly compatible with modern standard power supplies

Consequently, a number of other interface standards have been developed by the RS to overcome some of these limitations. The RS-485 interface standards are increasingly being used for instrumentation and control systems.

2.5 RS-232 troubleshooting

2.5.1 Introduction

Since RS-232 is a point-to-point system, installation is fairly straightforward and simple and all RS-232 devices use either DB-9 or DB-25 connectors. These connectors are used because they are cheap and allow multiple insertions. None of the 232 standards define which device uses a male or female connector, but traditionally the male (pin) connector is used on the DTE and the female type connector (socket) is used on DCE equipment. This is only traditional and may vary on different equipment. It is often asked why a 25-pin connector is used when only 9 pins are needed. This was done because RS-232 was used before the advent of computers. It was therefore used for hardware control (RTS/CTS). It was originally thought that, in the future, more hardware control lines would be needed hence the need for more pins.

When doing an initial installation of an RS-232 connection it is important to note the following:

- Is one device a DTE and the other a DCE?
- What is the sex and size of connectors at each end?
- What is the speed of the communication?
- What is the distance between the equipment?
- Is it a noisy environment?
- Is the software set up correctly?

2.6 Typical approach

When troubleshooting a serial data communications interface, one needs to adopt a logical approach in order to avoid frustration and wasting many hours. A procedure similar to that outlined below is recommended:

- Check the basic parameters. Are the baud rate, stop/start bits and parity set identically for both devices? These are sometimes set on DIP switches in the device. However, the trend is towards using software, configured from a terminal, to set these basic parameters
- Identify which is DTE or DCE. Examine the documentation to establish what actually happens at pins 2 and 3 of each device. On the 25 pin DTE device, pin 2 is used for transmission of data and should have a negative voltage (mark) in idle state, whilst pin 3 is used for the receipt of data (passive) and should be approximately at 0 Volts. Conversely, at the DCE device, pin 3 should have a negative voltage, whilst pin 2 should be around 0 Volts. If no voltage can be detected on either pin 2 or 3, then the device is probably not RS-232 compatible and could be connected according to another interface standard, such as RS-422, RS-485, etc

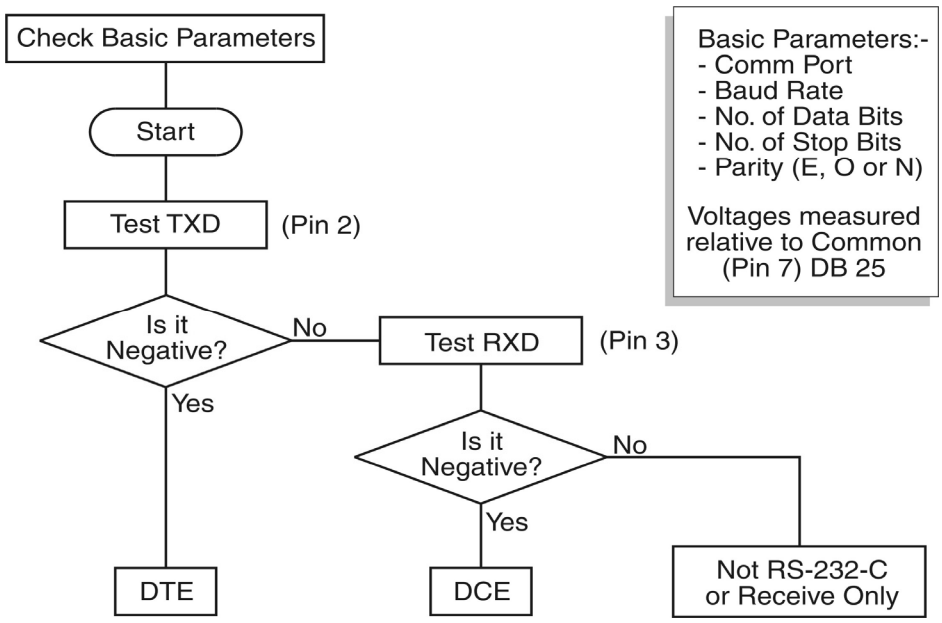


Figure 2.5
Flowchart to identify an RS-232 device as either a DTE or DCE

- Clarify the needs of the hardware handshaking when used. Hardware handshaking can cause the greatest difficulties and the documentation should be carefully studied to yield some clues about the handshaking sequence. Ensure all the required wires are correctly terminated in the cables
- Check the actual protocol used. This is seldom a problem but, when the above three points do not yield an answer, it is possible that there are irregularities in the protocol structure between the DCE and DTE devices
- Alternatively, if software handshaking is utilized, make sure both have compatible application software. In particular, check that the same ASCII character is used for XON and XOFF

2.7 Test equipment

From a testing point of view, the RS-232-E interface standard states that:

'The generator on the interchange circuit shall be designed to withstand an open circuit, a short circuit between the conductor carrying that interchange circuit in the interconnecting cable and any other conductor in that cable including signal ground, without sustaining damage to itself or its associated equipment.'

In other words, any pin may be connected to any other pin, or even earth, without damage and, theoretically, one cannot blow up anything! This does not mean that the RS-232 interface cannot be damaged. The incorrect connection of incompatible external voltages can damage the interface, as can static charges.

If a data communication link is inoperable, the following devices may be useful when analyzing the problem:

- A digital multimeter. Any cable breakage can be detected by measuring the continuity of the cable for each line. The voltages at the pins in active and inactive states can also be ascertained by the multimeter to verify its compatibility to the respective standards.
- An LED. The use of an LED is to determine which are the asserted lines or whether the interface conforms to a particular standard. This is laborious and accurate pin descriptions should be available.
- A breakout box
- PC-based protocol analyzer (including software)
- Dedicated hardware protocol analyzer (e.g. Hewlett Packard)

2.7.1 The breakout box

The breakout box is an inexpensive tool that provides most of the information necessary to identify and fix problems on data communications circuits, such as the serial RS-232, RS-422, RS-423 and RS-485 interfaces and also on parallel interfaces.

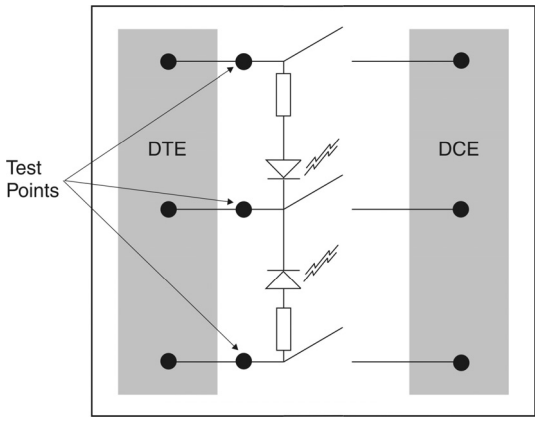


Figure 2.6
Breakout box showing test points

A breakout box is connected to the data cable, to bring out all conductors in the cable to accessible test points. Many versions of this equipment are available on the market, from the 'homemade' using a back-to-back pair of male and female DB-25 sockets to fairly sophisticated test units with built-in LEDs, switches and test points.

Breakout boxes usually have a male and a female socket and by using two standard serial cables, the box can be connected in series with the communication link. The 25 test points can be monitored by LEDs, a simple digital multimeter, an oscilloscope or a protocol analyzer. In addition, a switch in each line can be opened or closed while trying to identify the problem.

The major weakness of the breakout box is that while one can interrupt any of the data lines, it does not help much with the interpretation of the flow of bits on the data communication lines. A protocol analyzer is required for this purpose.

2.7.2 Null modem

Null modems look like DB-25 'through' connectors and are used when interfacing two devices of the same gender (e.g. DTE to DTE, DCE to DCE) or devices from different manufacturers with different handshaking requirements. A null modem has appropriate internal connections between handshaking pins that 'trick' the terminal into believing conditions are correct for passing data. A similar result can be achieved by soldering extra loops inside the DB-25 plug. Null modems generally cause more problems than they cure and should be used with extreme caution and preferably avoided.

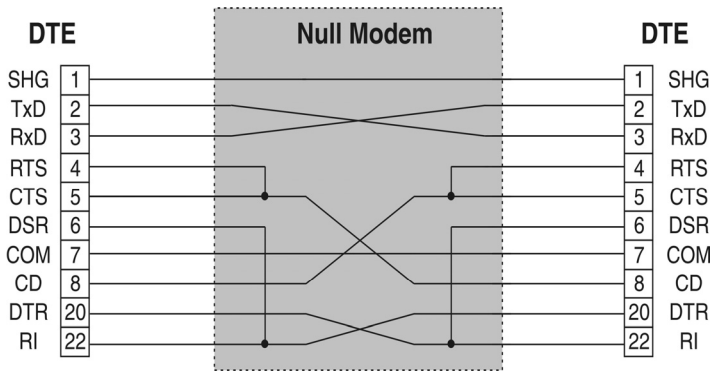


Figure 2.7
Null modem connections

Note that the null modem may inadvertently connect pins 1 together, as in Figure 2.7. This is an undesirable practice and should be avoided.

2.7.3 Loop back plug

This is a hardware plug which loops back the transmit data pin to receive data pin and similarly for the hardware handshaking lines. This is another quick way of verifying the operation of the serial interface without connecting to another system.

2.7.4 Protocol analyzer

A protocol analyzer is used to display the actual bits on the data line, as well as the special control codes, such as STX, DLE, LF, CR, etc. The protocol analyzer can be used to monitor the data bits, as they are sent down the line and compared with what should be on the line. This helps to confirm that the transmitting terminal is sending the correct data and that the receiving device is receiving it. The protocol analyzer is useful in identifying incorrect baud rate, incorrect parity generation method, incorrect number of stop bits, noise, or incorrect wiring and connection. It also makes it possible to analyze the format of the message and look for protocol errors.

When the problem has been shown not to be due to the connections, baud rate, bits or parity, then the content of the message will have to be analyzed for errors or inconsistencies. Protocol analyzers can quickly identify these problems.

Purpose built protocol analyzers are expensive devices and it is often difficult to justify the cost when it is unlikely that the unit will be used very often. Fortunately, software has been developed that enables a normal PC to be used as a protocol analyzer. The use of a PC as a test device for many applications is a growing field, and one way of connecting a PC as a protocol analyzer is shown in Figure 2.8.

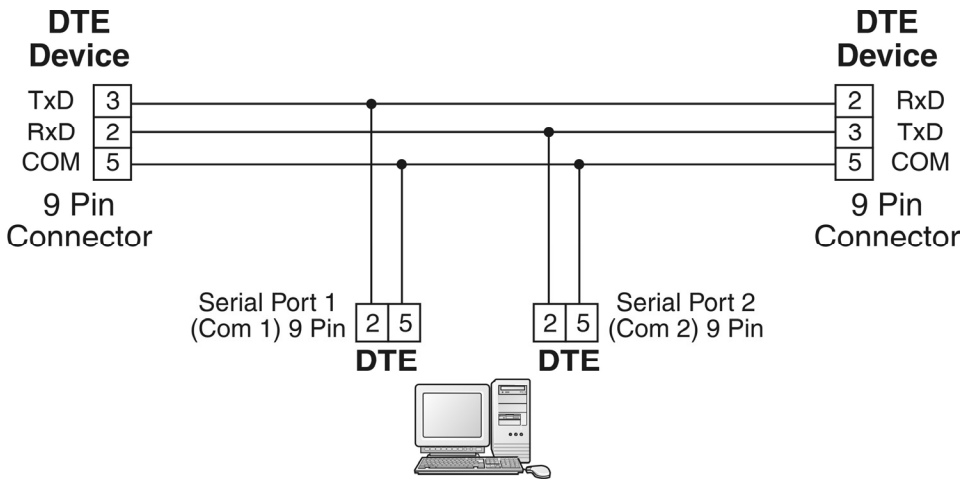


Figure 2.8
Protocol analyzer connection

The above figure has been simplified for clarity and does not show the connections on the control lines (For Example, RTS and CTS).

2.8 Typical RS-232 problems

Below is a list of typical RS-232 problems, which can arise because of inadequate interfacing. These problems could equally apply to two PCs connected to each other or to a PC connected to a printer.

Problem	Probable cause of problem
Garbled or lost data	Baud rates of connecting ports may be different
	Connecting cables could be defective
	Data formats may be inconsistent (Stop bit/ parity/ number of data bits)
	Flow control may be inadequate
	High error rate due to electrical interference
	Buffer size of receiver inadequate
First characters garbled	The receiving port may not be able to respond quickly enough. Precede the first few characters with the ASCII (DEL) code to ensure frame synchronization.
No data communications	Power for both devices may not be on
	Transmit and receive lines of cabling may be incorrect
	Handshaking lines of cabling may be incorrectly connected
	Baud rates mismatch
	Data format may be inconsistent
	Earth loop may have formed for RS-232 line
	Extremely high error rate due to electrical interference for transmitter and receiver
	Protocols may be inconsistent/ Intermittent communications
	Intermittent interference on cable
ASCII data has incorrect spacing	Mismatch between 'LF' and 'CR' characters generated by transmitting device and expected by receiving device.

Table 2.4
A list of typical RS-232 problems

To determine whether the devices are DTE or DCE, connect a breakout box at one end and note the condition of the TX light (pin 2 or 3) on the box. If pin 2 is ON, then the device is probably a DTE. If pin 3 is ON, it is probably a DCE. Another clue could be the sex of the connector, male are typically DTEs and females are typically DCEs, but not always.

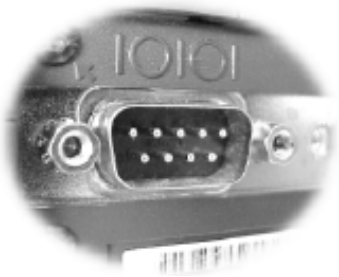


Figure 2.9
A 9 pin RS-232 connector on a DTE

When troubleshooting an RS-232 system, it is important to understand that there are two different approaches. One approach is followed if the system is new and never been run before and the other if the system has been operating and for some reason does not

communicate at present. New systems that have never worked have more potential problems than a system that has been working before and now has stopped. If a system is new it can have three main problems viz. mechanical, setup or noise. A previously working system usually has only one problem, viz. mechanical. This assumes that no one has changed the setup and noise has not been introduced into the system. In all systems, whether having previously worked or not, it is best to check the mechanical parts first.

This is done by:

- Verifying that there is power to the equipment
- Verifying that the connectors are not loose
- Verifying that the wires are correctly connected
- Checking that a part, board or module has not visibly failed

2.8.1 Mechanical problems

Often, mechanical problems develop in RS-232 systems because of incorrect installation of wires in the D-type connector or because strain reliefs were not installed correctly.

The following recommendations should be noted when building or installing RS-232 cables:

- Keep the wires short (20 meters maximum)
- Stranded wire should be used instead of solid wire (solid wire will not flex.)
- Only one wire should be soldered in each pin of the connector.
- Bare wire should not be showing out of the pin of the connector
- The back shell should reliably and properly secure the wire

The speed and distance of the equipment will determine if it is possible to make the connection at all. Most engineers try to stay less than 50 feet or about 16 meters at 115200 bits per second. This is a very subjective measurement and will depend on the cable, voltage of the transmitter and the amount and noise in the environment. The transmitter voltage can be measured at each end when the cable has been installed. A voltage of at least +/– 5 V should be measured at each end on both the TX and RX lines.

An RS-232 breakout box is placed between the DTE and DCE to monitor the voltages placed on the wires by looking at pin 2 on the breakout box. Be careful here because it is possible that the data is being transmitted so fast that the light on the breakout box doesn't have time to change. If possible, lower the speed of the communication at both ends to something like 2 bps.

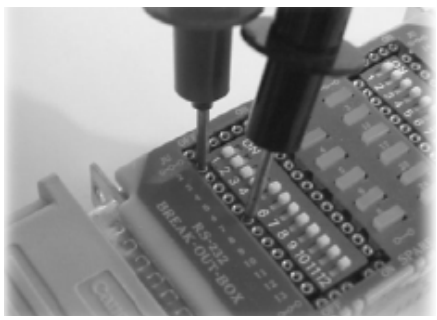


Figure 2.10
Measuring the voltage on RS-232

Once it has been determined that the wires are connected as DTE to DCE and that the distance and speed are not going to be a problem, the cable can be connected at each end. The breakout box can still be left connected with the cable and both pin 2 and 3 lights on the breakout box should now be on.

The color of the light depends on the breakout box. Some breakout boxes use red for a one and others use green for a one. If only one light is on then that may mean that a wire is broken or there is a DTE to DTE connection. A clue to a possible DTE to DTE connection would be that the light on pin 3 would be off and the one on pin 2 would be on. To correct this problem, first check the wires for continuity then turn switches 2 and 3 off on the breakout box and use jumper wires to swap them. If the TX and RX lights come on, a null modem cable or box will need to be built and inserted in-line with the cable.

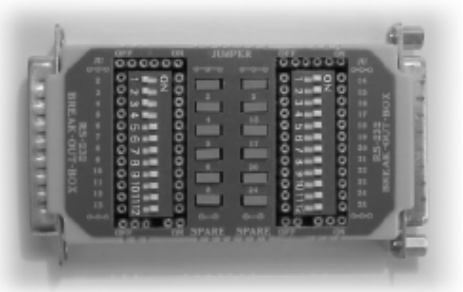


Figure 2.11
A RS-232 breakout box

If the pin 2 and pin 3 lights are on, one end is transmitting and the control is correct, then the only thing left is the protocol or noise. Either a hardware or software protocol analyzer will be needed to troubleshoot the communications between the devices. On new installations, one common problem is mismatched baud rates. The protocol analyzer will tell exactly what the baud rates are for each device. Another thing to look for with the analyzer is the timing. Often, the transmitter waits some time before expecting a proper response from the receiver. If the receiver takes too long to respond or the response is incorrect, the transmitter will 'time out.' This is usually denoted as a ‘communications error or failure.’

2.8.2 Setup problems

Once it is determined that the cable is connected correctly and the proper voltage is being received at each end, it is time to check the setup. The following circumstances need to be checked before trying to communicate:

- Is the software communications setup at both ends for either 8N1, 7E1 or 7O1?
- Is the baud rate the same at both devices? (1200, 4800, 9600, 19200 etc.)
- Is the software setup at both ends for binary, hex or ASCII data transfer?
- Is the software setup for the proper type of control?

Although the 8 data bits, no parity and 1 stop bit is the most common setup for asynchronous communication, often 7 data bits even parity with 1 stop bit is used in industrial equipment. The most common baud rate used in asynchronous communications is 9600. Hex and ASCII are commonly used as communication codes.

If one device is transmitting but the other receiver is not responding, then the next thing to look for is what type of control the devices are using. The equipment manual may define whether hardware or software control is being used. Both ends should be set up either for hardware control, software control or none.

2.8.3 Noise problems

RS-232, being a single ended (unbalanced) type of circuit, lends itself to receiving noise. There are three ways that noise can be induced into an RS-232 circuit.

- Induced noise on the common ground
- Induced noise on the TX or RX lines
- Induced noise on the indicator or control lines

Ground induced noise

Different ground voltage levels on the ground line (pin7) can cause ground loop noise. Also, varying voltage levels induced on the ground at either end by high power equipment can cause intermittent noise. This kind of noise can be very difficult to reduce. Sometimes, changing the location of the ground on either the RS-232 equipment or the high power equipment can help, but this is often not possible. If it is determined that the noise problem is caused by the ground it may be best to replace the RS-232 link with a fiber optic or RS-422 system. Fiber optic or RS-422 to RS-232 adapters are relatively cheap, readily available and easy to install. When the cost of troubleshooting the system is included, replacing the system often is the cheapest option.

Induced noise on the TX or RX lines

Noise from the outside can cause the communication on an RS-232 system to fail, although this voltage must be quite large. Because RS-232 voltages in practice are usually between +/– 7 and +/– 12, the noise voltage value must be quite high in order to induce errors. This type of noise induction is noticeable because the voltage on the TX or RX will be outside of the specifications of RS-232. Noise on the TX line can also be induced on the RX line (or *vice versa*) due to the common ground in the circuit. This type of noise can be detected by comparing the data being transmitted with the received communication at the other end of the wire (assuming no broken wire). The protocol analyzer is plugged into the transmitter at one end and the data monitored. If the data is correct, the protocol analyzer is then plugged into the other end and the received data monitored. If the data is corrupt at the receiving end, then noise on that wire may be the problem. If it is determined that the noise problem is caused by induced noise on the TX or RX lines, it may be best to move the RS-232 line and the offending noise source away from each other. If this doesn't help, it may be necessary to replace the RS-232 link with a fiber optic or RS-485 system.

Induced noise on the indicator or control lines

This type of noise is very similar to the previous TX/RX noise. The difference is that noise on these wires may be harder to find. This is because the data is being received at both ends, but there still is a communication problem. The use of a voltmeter or oscilloscope would help to measure the voltage on the control or indicator lines and therefore locate the possible cause of the problem, although this is not always very accurate. This is because the effect of noise on a system is governed by the ratio of the power levels of the signal and the noise, rather than a ratio of their respective voltage levels. If it is determined that the noise is being induced on one of the indicator or control

lines, it may be best to move the RS-232 line and the offending noise source away from each other. If this doesn't help, it may be necessary to replace the RS-232 link with a fiber optic or RS-485 system.

2.9 Summary of troubleshooting

Installation

- Is one device a DTE and the other a DCE?
- What is the sex and size of the connectors at each end?
- What is the speed of the communications?
- What is the distance between the equipment?
- Is it a noisy environment?
- Is the software set up correctly?

Troubleshooting new and old systems

- Verify that there is power to the equipment
- Verify that the connectors are not loose
- Verify that the wires are correctly connected
- Check that a part, board or module has not visibly failed

Mechanical problems on new systems

- Keep the wires short (20 meters maximum)
- Stranded wire should be used instead of solid wire (stranded wire will flex)
- Only one wire should be soldered in each pin of the connector
- Bare wire should not be showing out of the connector pins
- The back shell should reliably and properly secure the wire

Setup problems on new systems

- Is the software communications set up at both ends for either 8N1, 7E1 or 7O1?
- Is the baud rate the same for both devices? (1200,4800,9600,19200 etc.)
- Is the software set up at both ends for binary, hex or ASCII data transfer?
- Is the software setup for the proper type of control?

Noise problems on new systems

- Noise from the common ground
- Induced noise on the TX or RX lines
- Induced noise on the indicator or control lines